Fundamentals of Cloud Computing

Final Project

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Introduction

This module covers cloud computing fundamentals with various platforms such as AWS, Google Cloud, and Microsoft Azure with the intent of preparing for the CompTIA Cloud+ examination.

The focus was on the process of building a system in Microsoft Azure which consisted of Virtual Machine (VM) deployment, Virtual Network (VNet) configurations, VM security, cloud storage, and cloud monitoring.

This journey demonstrates hands-on experience working with cloud computing and professional knowledge of its application in real-world scenarios

Virtual Machine (VM) Instances

Virtual Machine (VM) Instances

This part of the module explored how to create, test, and delete a virtual machine instance.

The initial step was to deploy a new a virtual machine within the Microsoft Azure virtual machine (VM) service. This consisted of selecting the correct subscription, region, image, and size. Naming the machine, creating a username and establishing a password were also involved in this process.

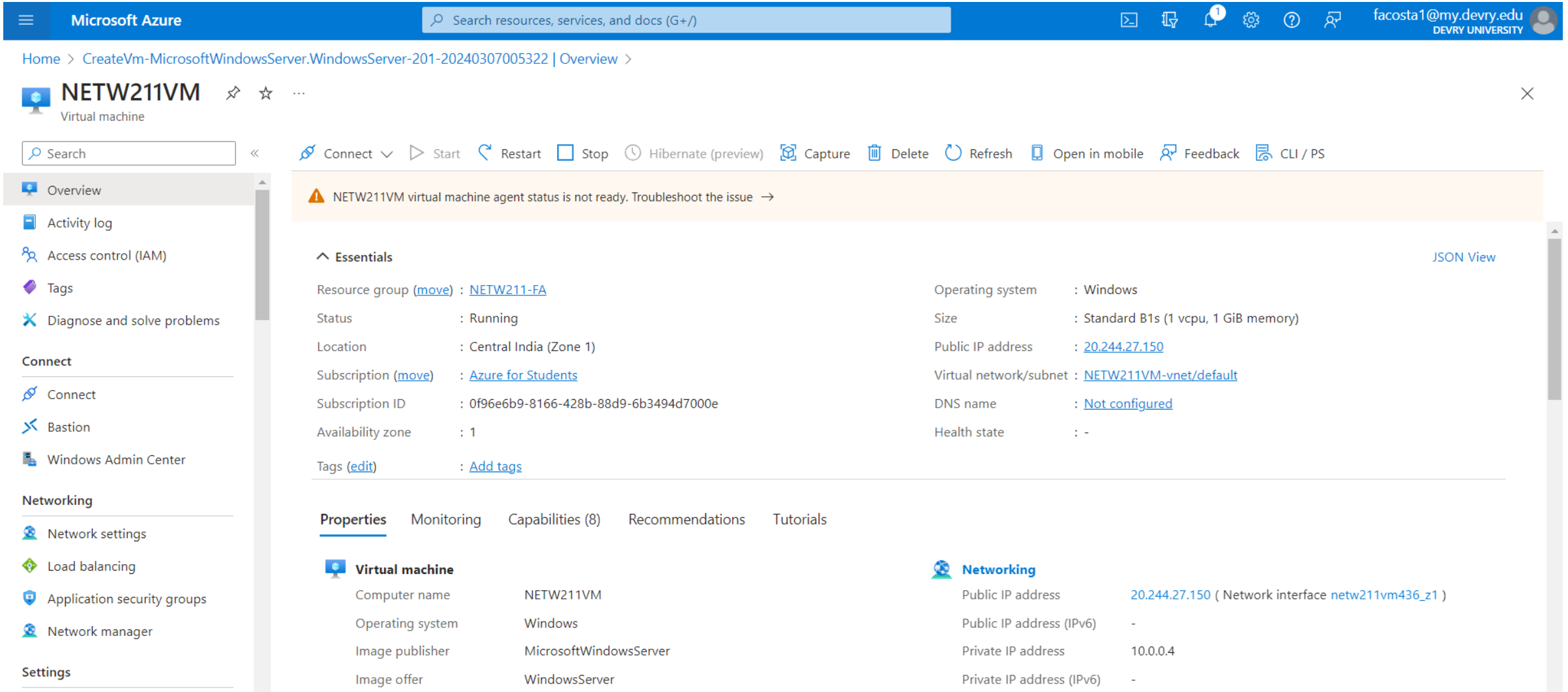
The next step was to test the VM's connection via the Windows Remote Desktop Protocol (RDP).

The local server was then reviewed to verify the VM is correctly set to the intended specifications.

The deletion of the VM and resource group within Azure was the last step taken to complete this portion of the module.

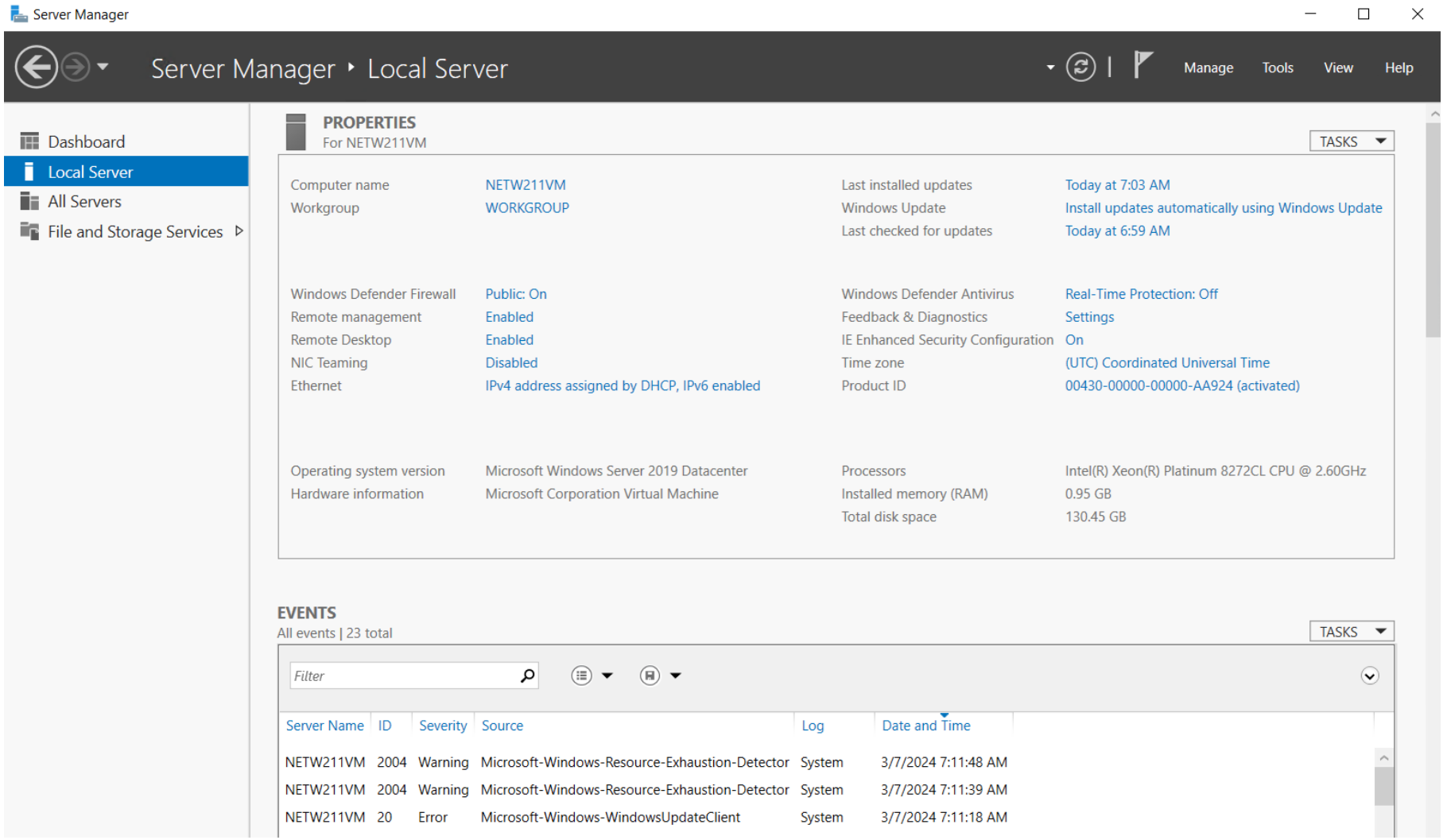
Deploying a VM in Microsoft Azure

VM Overview



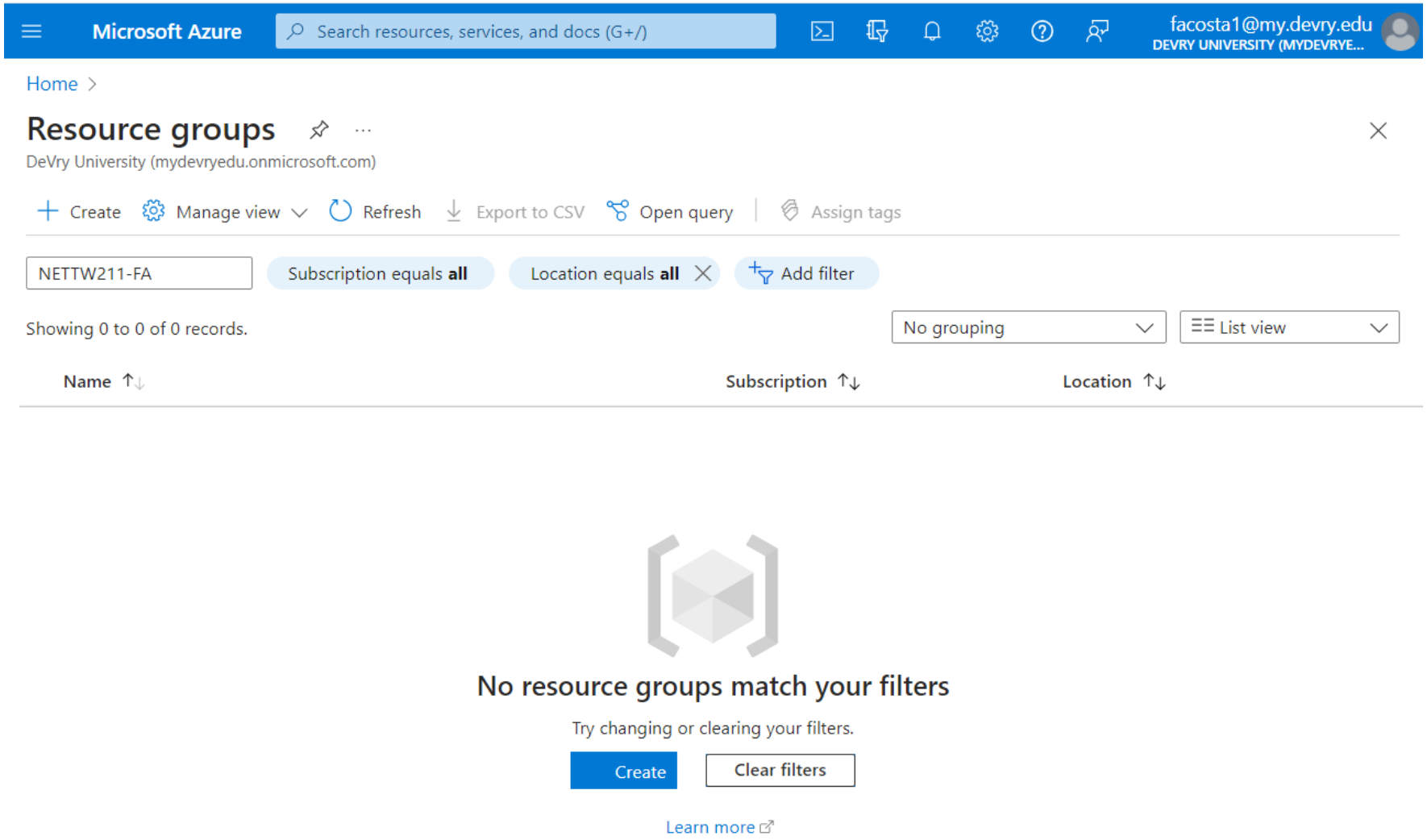
Connecting to the VM

VM Properties



Deleting the VM

Reviewing the Resource Groups



Azure VNet and Subnets

Azure VNet and Subnets

This section of the module consisted of creating a virtual network with two subnets within Microsoft Azure.

The virtual network was configured as a class C network with a /24 subnet mask.

Each virtual machine was imaged with Microsoft Windows Server 2019 with a standard B1s size.

Testing was vital part of this project because it ensured each machine was able to connect to one another.

Creating a VNet with Two Subnets

1. With a /24 network prefix, how many usable IPv4 host addresses are there?

There are 254 available IPv4 host addresses.

The network starts with 256 but two are subtracted for the network and broadcast addresses.

2. Given the answer above, why is the number of available IP addresses for Subnet0 (10.0.0.0/24) or Subnet1 (10.0.1.0/24) is 251 when you create an Azure VNet?

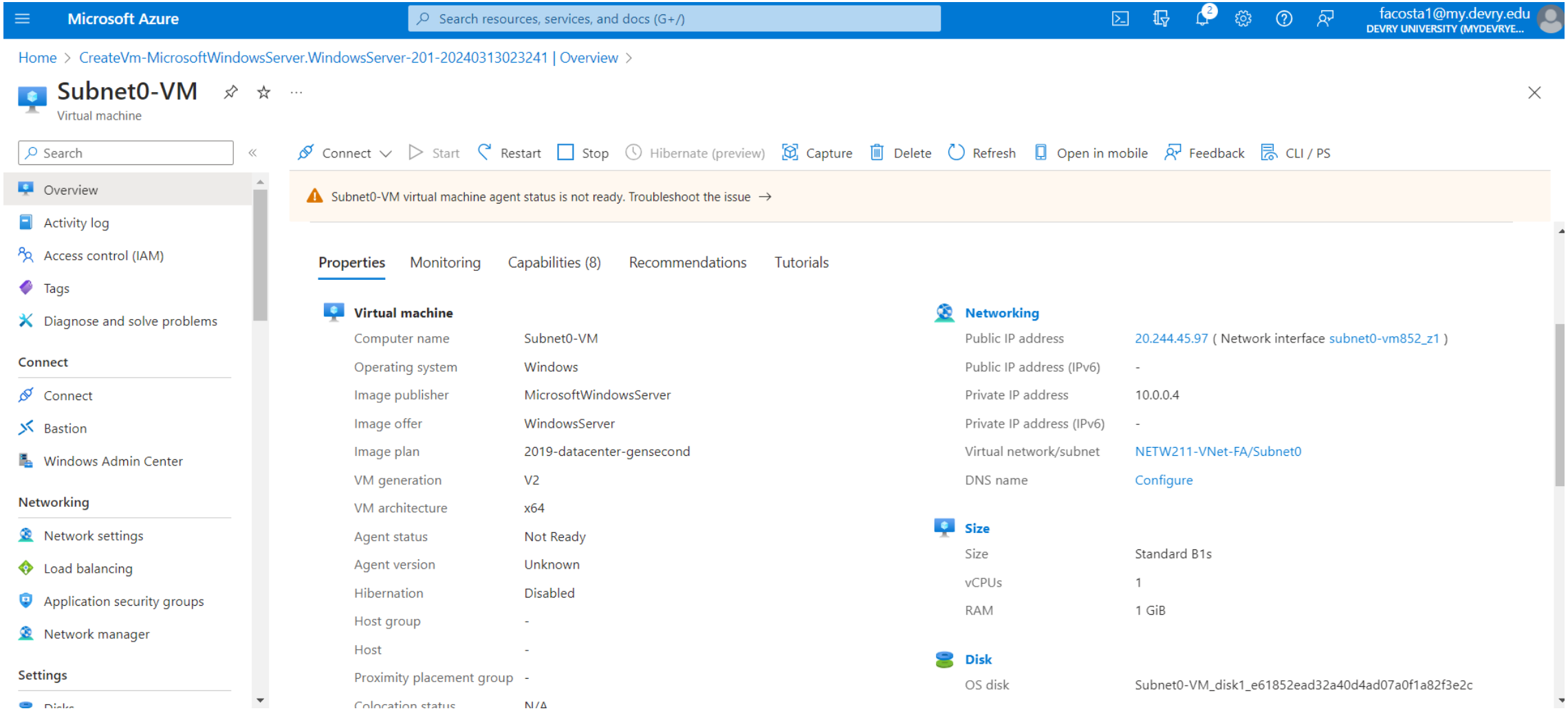
Azure reserves two addresses for the network and broadcast addresses but also requires three additional addresses for virtualization.

These additional reservations are for the VPC router, DNS, and Azure services.

A class C /24 Azure VNet network has 256 host addresses minus five reserved IP addresses which leaves 251 available.

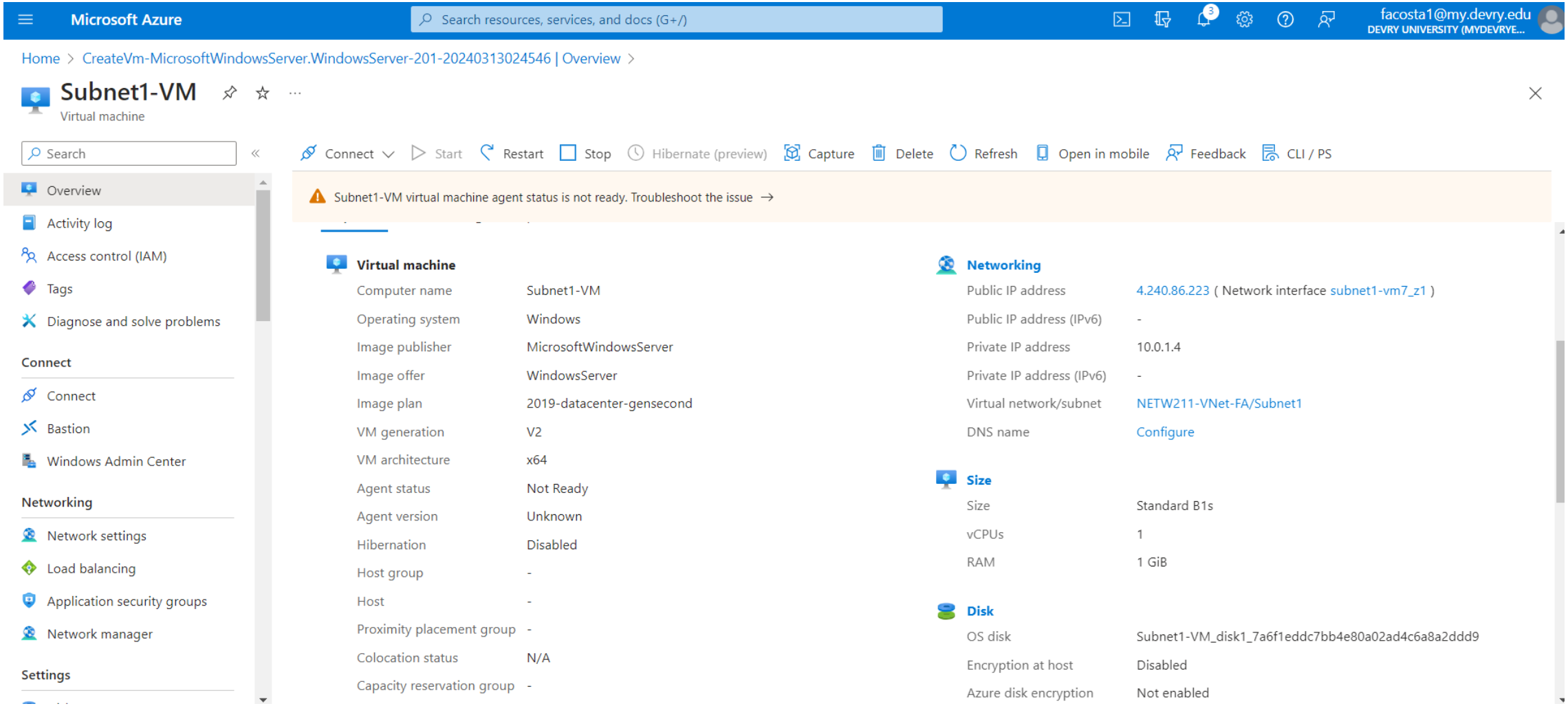
Deploying VMs into Subnets (1 of 3)

Properties section of the Subnet0-VM



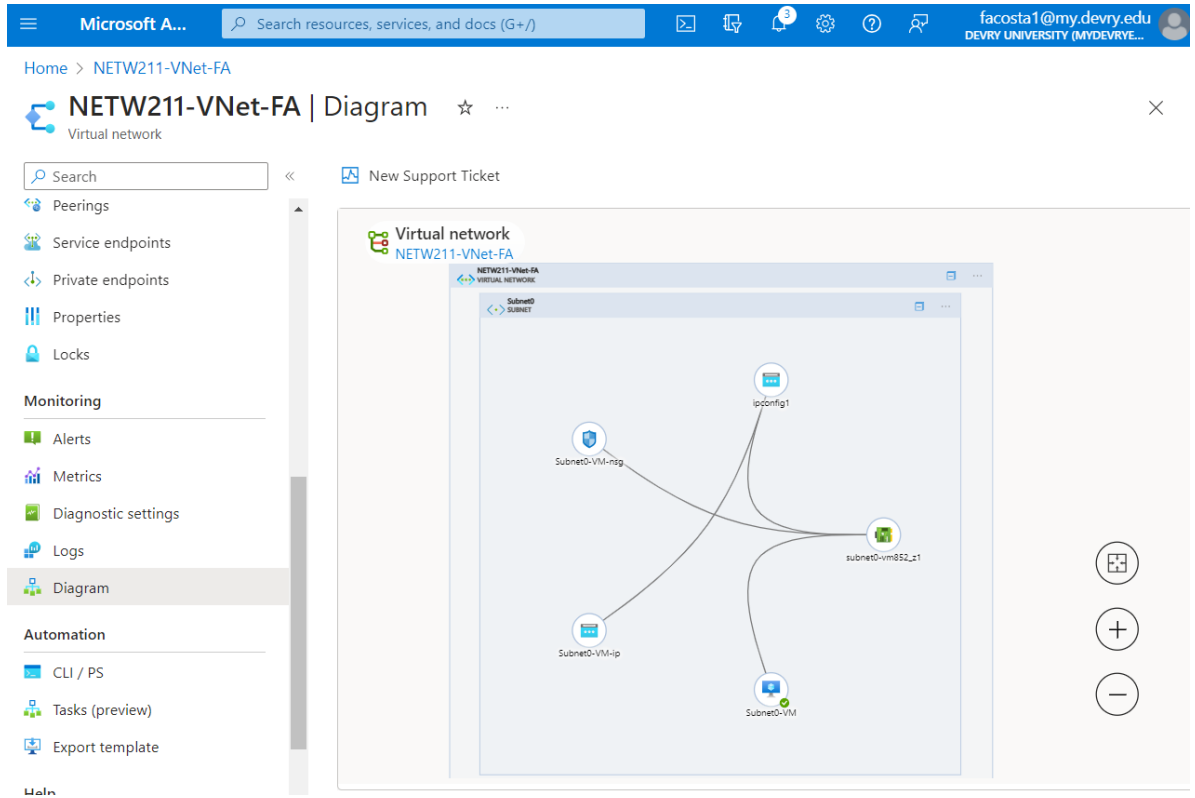
Deploying VMs into Subnets (2 of 3)

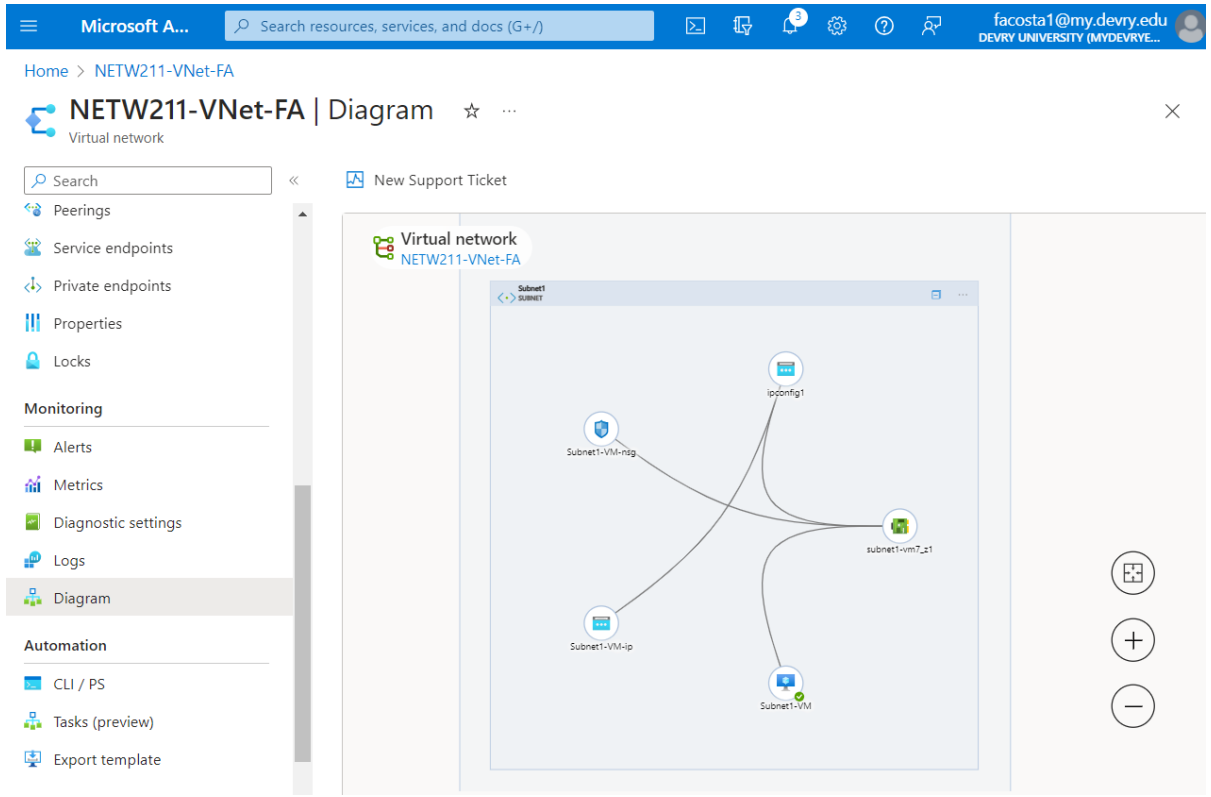
Properties section of the Subnet1-VM



Deploying VMs into Subnets (3 of 3)

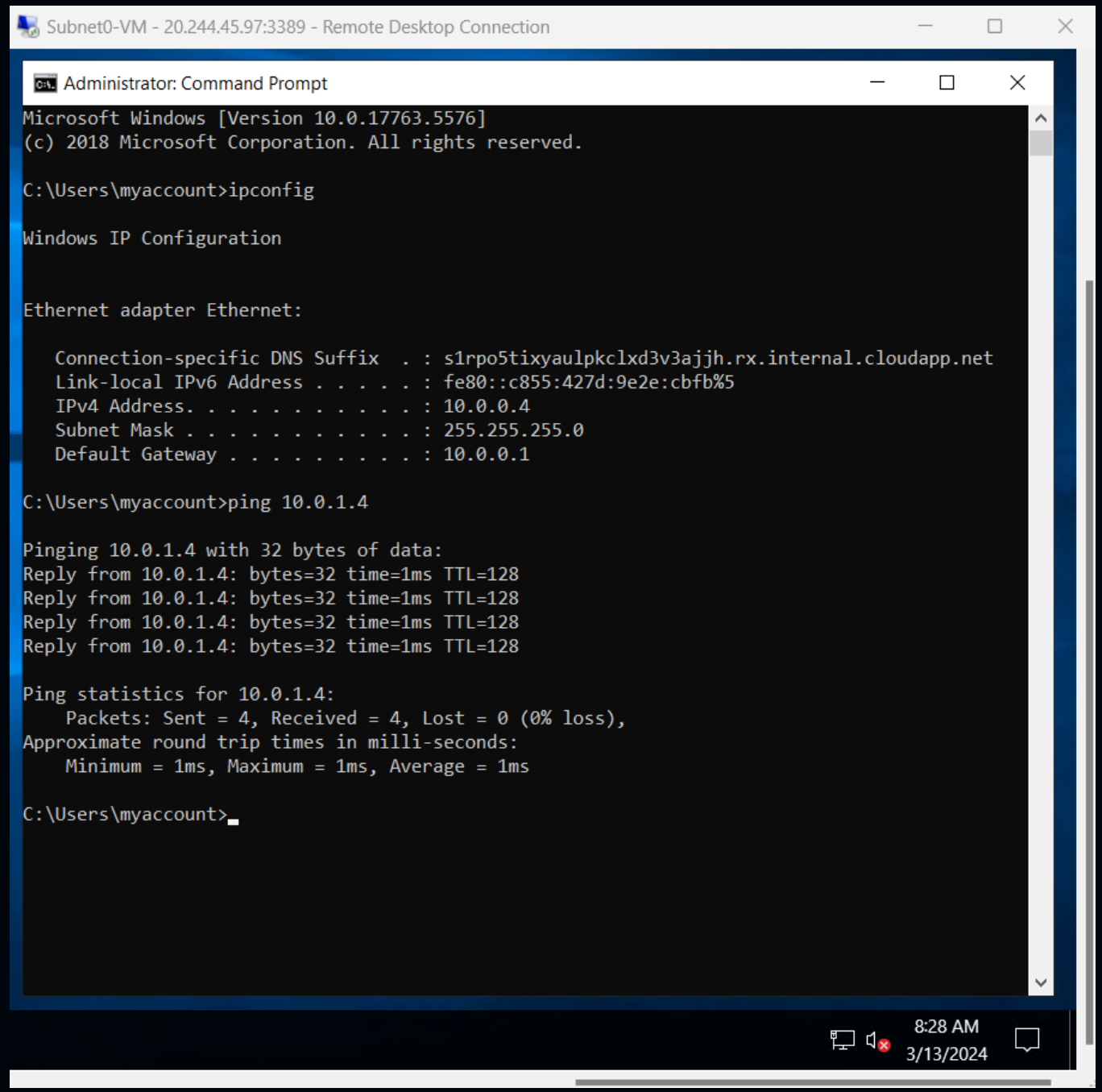
VNet Diagram





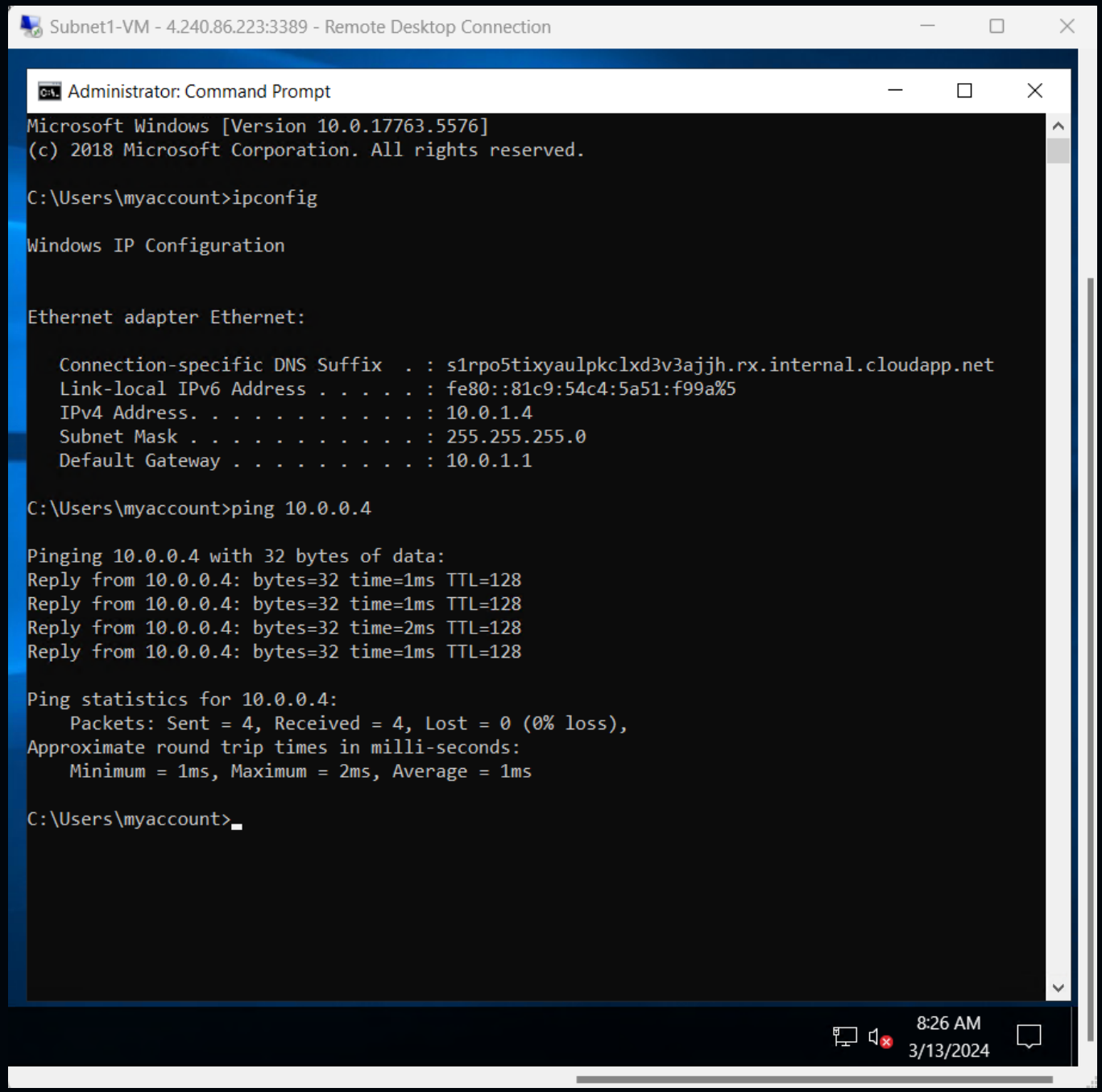
Verifying Connectivity between VMs (1 of 2)

Connection from VM 0 to VM 1



Verifying Connectivity between VMs (2 of 2)

Connection from VM 1 to VM 0



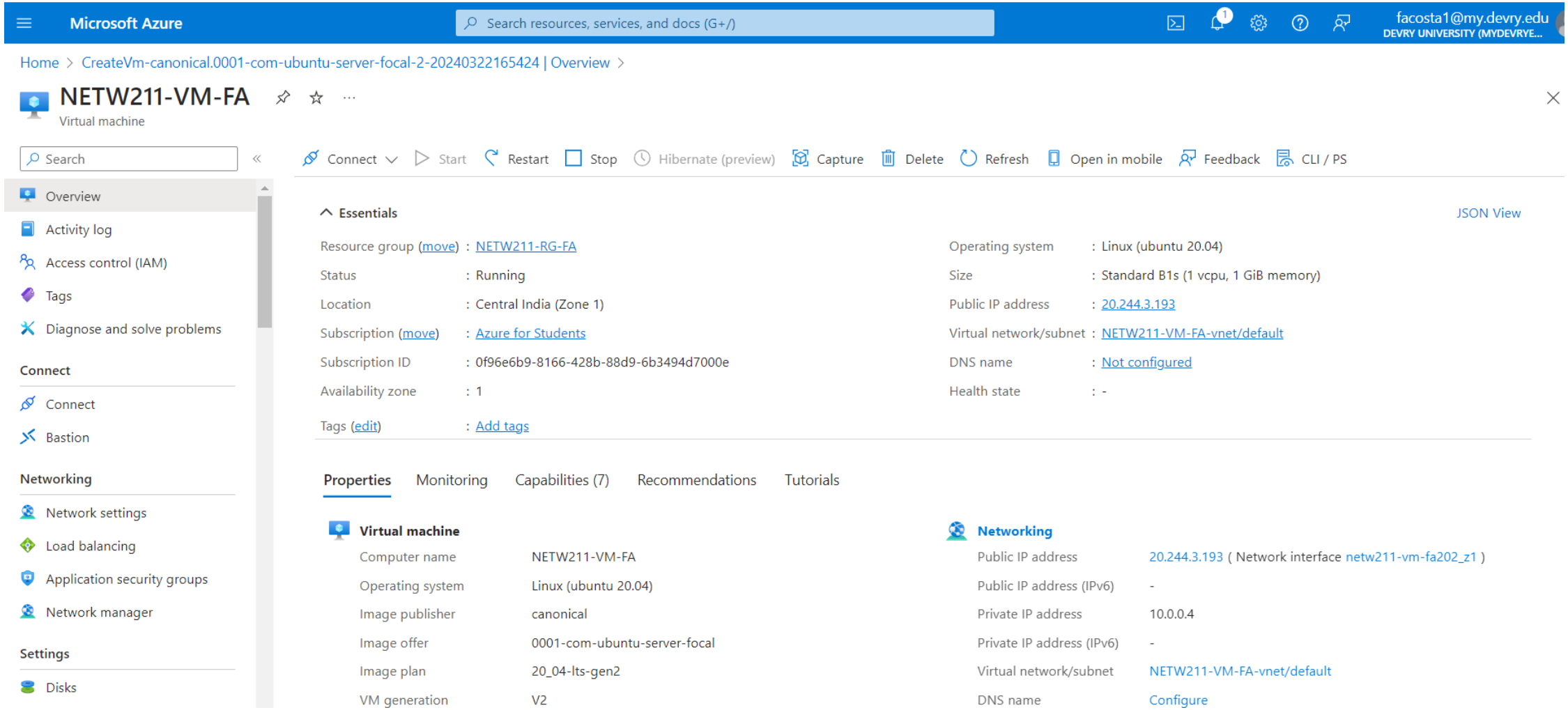
Azure VM Security

Azure VM Security

This portion of the module illustrates my ability to access a virtual machine via the Secure Shell (SSH) protocol and properly configure a network security group with the intent of controlling inbound and outbound traffic.

Major steps include the creation of a virtual machine within Microsoft Azure, the utilization of the Windows command prompt to connect to a virtual machine via SSH, and the configuration of a network security group to allow the flow of traffic.

Launching a VM

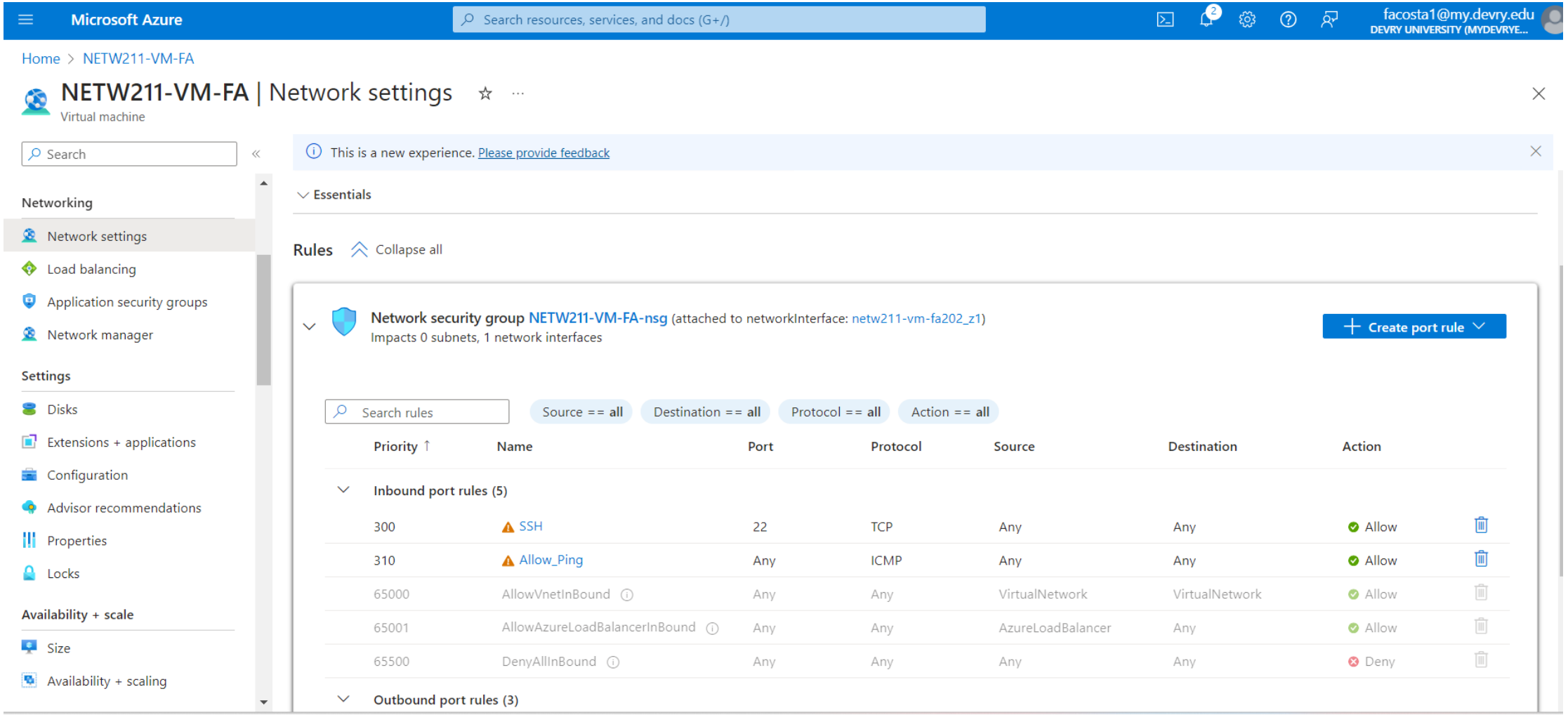


Connecting to the VM via SSH

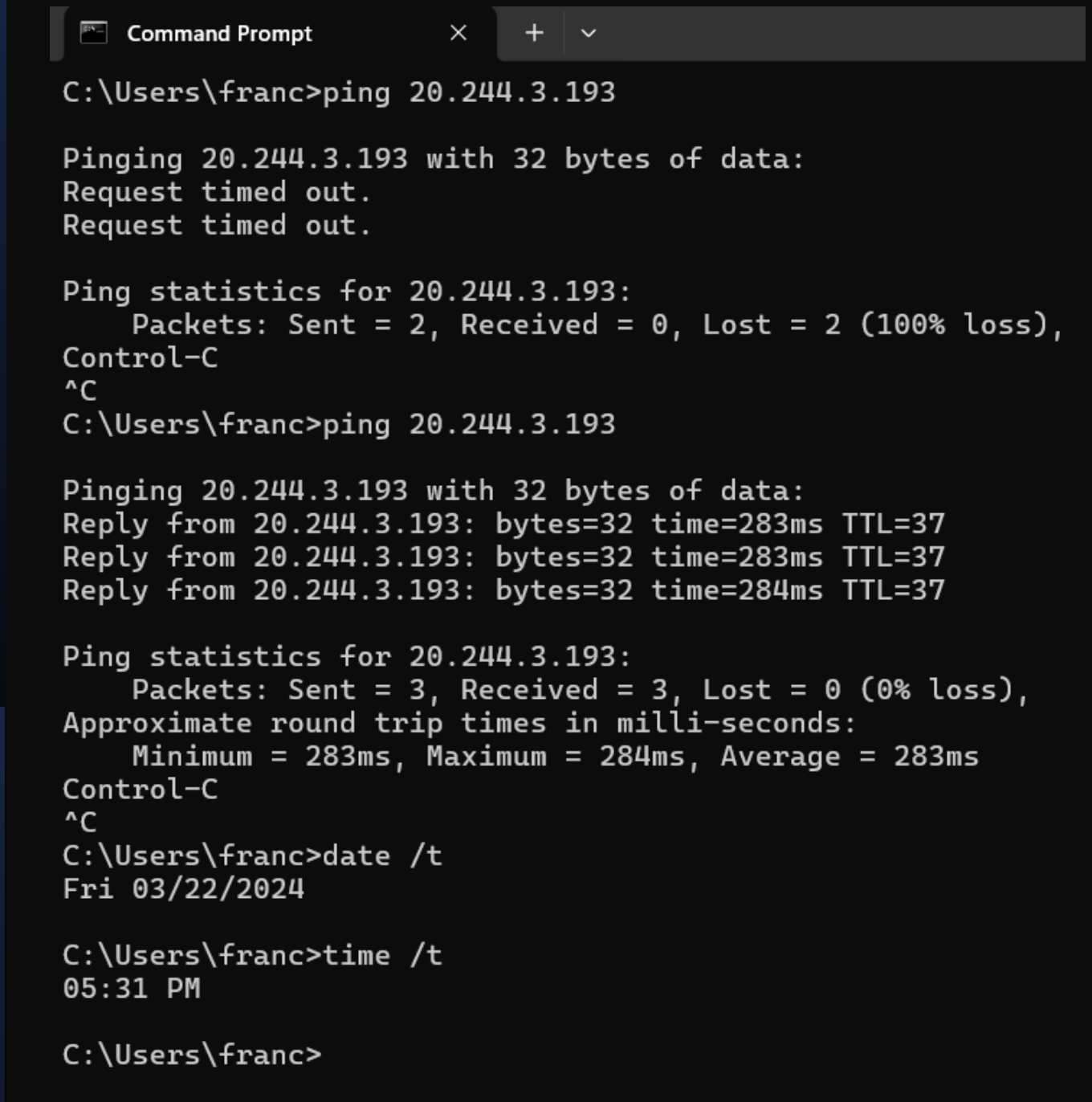
azureuser@NETW211-VM-FA:

```
azureuser@NETW211-VM-FA:~$ uname -r
5.15.0-1059-azure
azureuser@NETW211-VM-FA:~$ cat /etc/os-release
NAME="Ubuntu"
VERSION="20.04.6 LTS (Focal Fossa)"
ID=ubuntu
ID_LIKE=debian
PRETTY_NAME="Ubuntu 20.04.6 LTS"
VERSION_ID="20.04"
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-polic
VERSION_CODENAME=focal
UBUNTU_CODENAME=focal
azureuser@NETW211-VM-FA:~$ ping -c 4 www.facebook.com
PING star-mini.c10r.facebook.com (157.240.13.35) 56(84) bytes of data.
64 bytes from edge-star-mini-shv-02-sin6.facebook.com (157.240.13.35): icmp_seq=1
64 bytes from edge-star-mini-shv-02-sin6.facebook.com (157.240.13.35): icmp_seq=2
64 bytes from edge-star-mini-shv-02-sin6.facebook.com (157.240.13.35): icmp_seq=3
^C
--- star-mini.c10r.facebook.com ping statistics ---
4 packets transmitted, 3 received, 25% packet loss, time 3005ms
rtt min/avg/max/mdev = 47.458/48.395/50.106/1.211 ms
azureuser@NETW211-VM-FA:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc mq state UP group defau
    link/ether 60:45:bd:ce:db:f0 brd ff:ff:ff:ff:ff:ff
    inet 10.0.0.4/24 brd 10.0.0.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 fe80::6245:bdff:fece:dbf0/64 scope link
       valid_lft forever preferred_lft forever
azureuser@NETW211-VM-FA:~$ date
Fri Mar 22 22:17:23 UTC 2024
azureuser@NETW211-VM-FA:~$
```

Configuring a NSG 1 of 2



Configuring a NSG 2 of 2



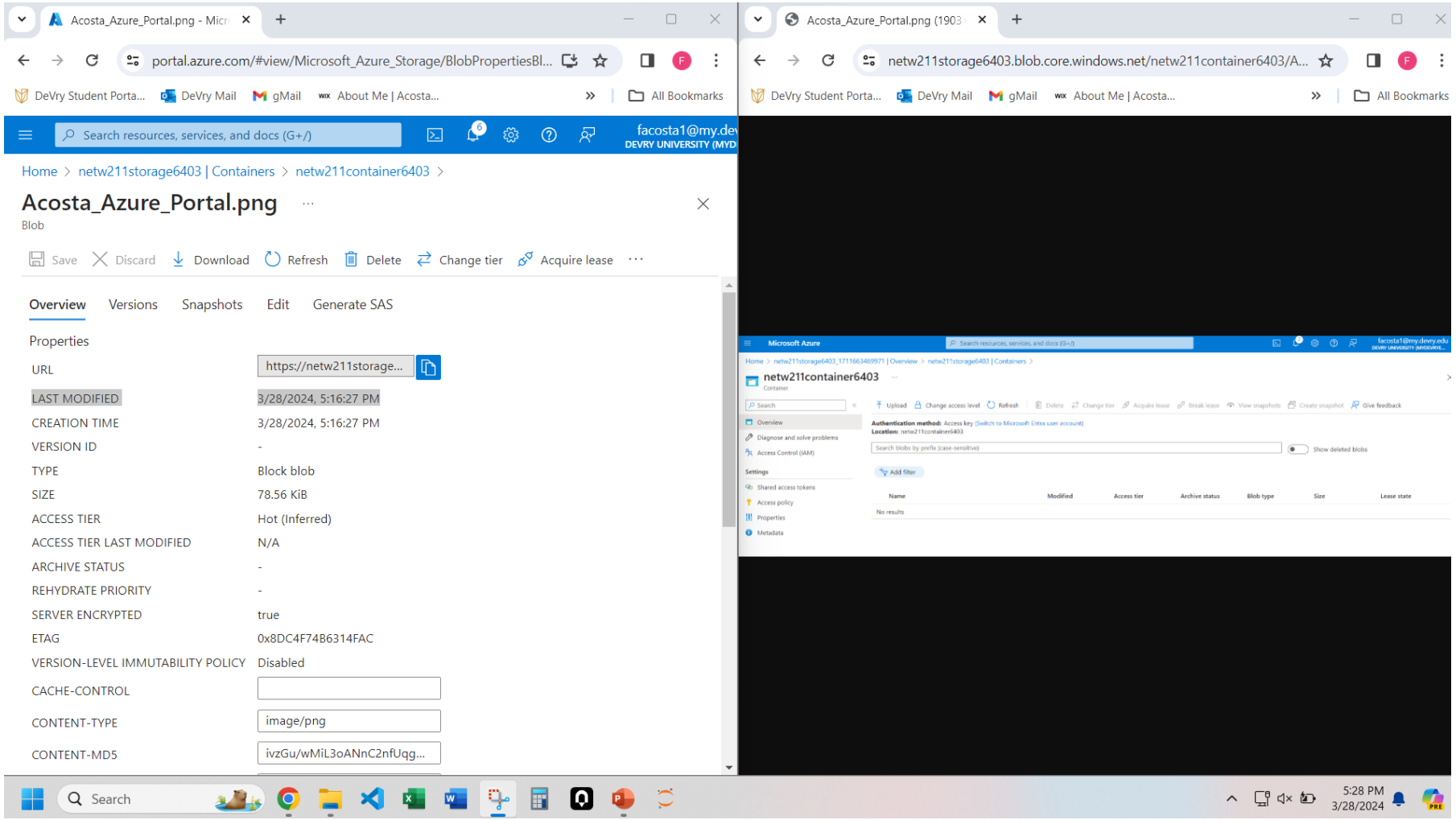
Cloud Storage

Cloud Storage

This section of the module illustrates my venture of exploring Microsoft Azure's Blob storage system. This storage system is an object-based cloud solution that optimizes the storage of large amounts of unstructured data that may include documents, videos, audio files, logs, archives, backups, and disaster recovery material.

Major steps included the creation of a storage account, establishment of a container, uploading a file, accessing a file, creation of blob snapshots, and enabling blob versioning.

Uploading and Accessing a File

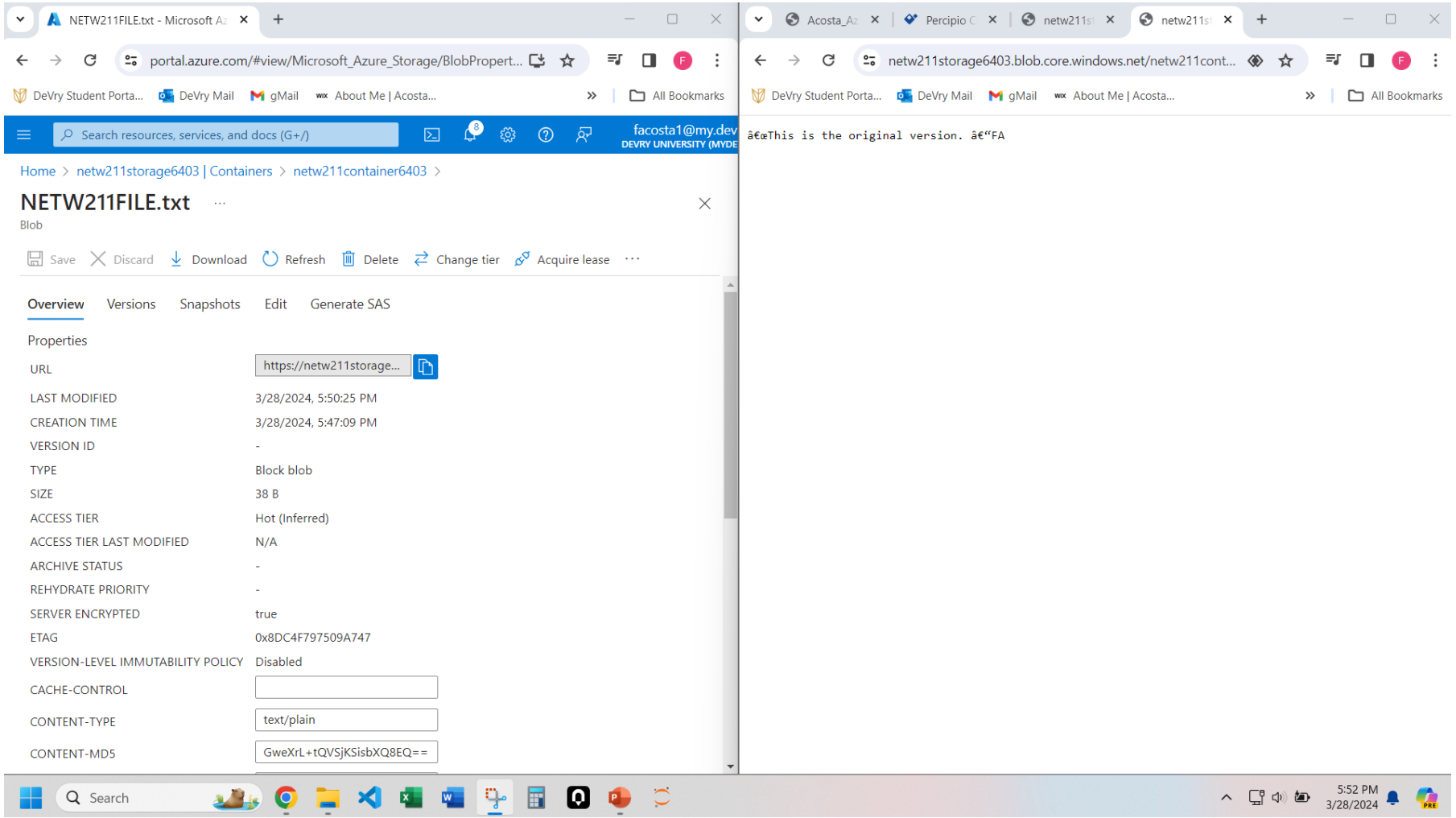


Creating Blob Snapshots 1 of 2

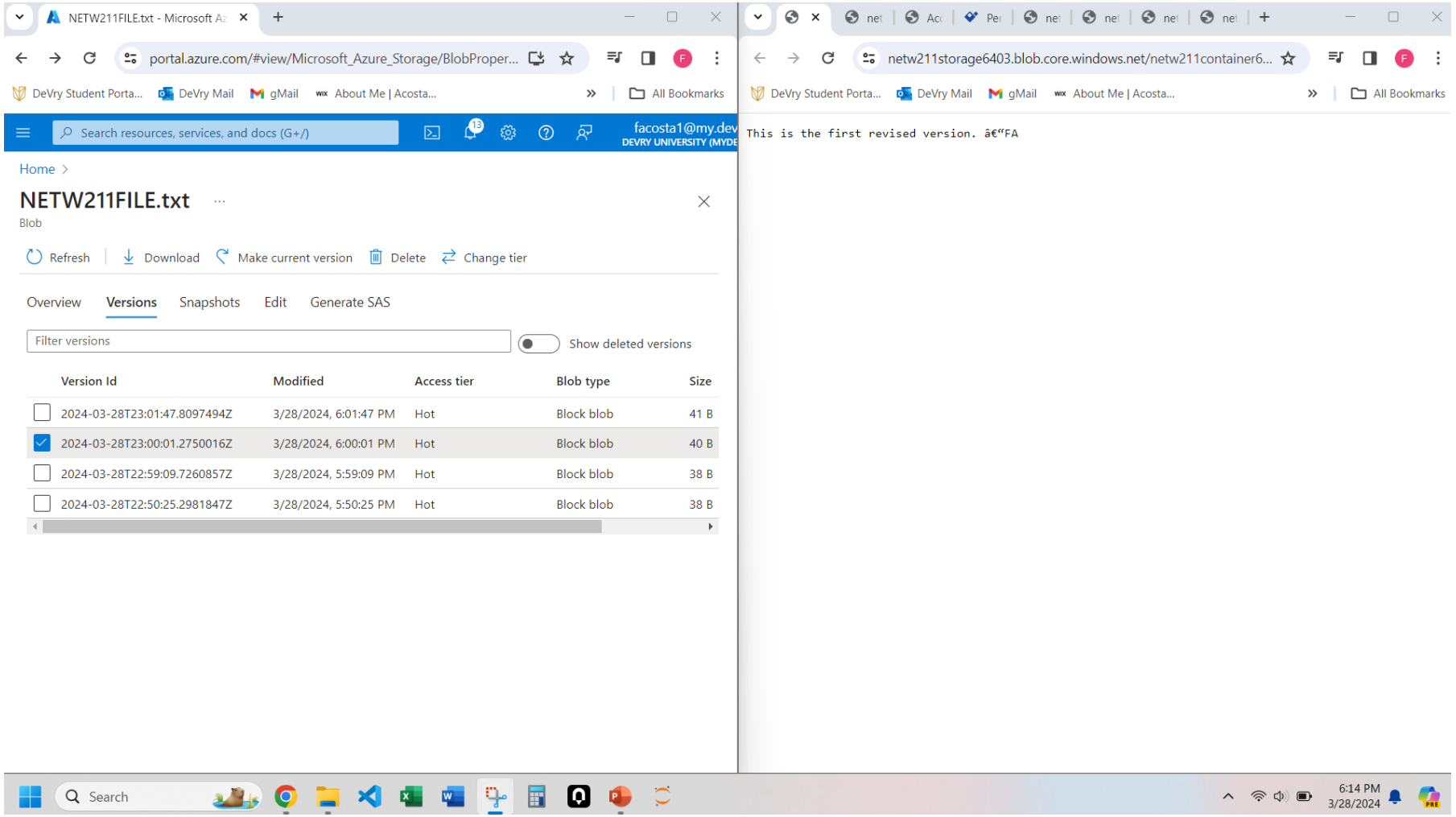
What does the access tier setting do?
Access tiering **optimizes** the **cost** of expanding storage needs by organizing data based on the frequency it is accessed and how long it is retained.

What are the Azure blob storage access tiers?
- The **Hot Tier** is for storing data that is accessed or modified frequently.
- The **Cool Tier** is for storing data that is infrequently accessed or modified.
- The **Cold Tier** is for storing data that is seldomly accessed or modified.
- The **Archive Tier** is for storing data that is rarely accessed.

Creating Blob Snapshots 2 of 2



Enabling Blob Versioning



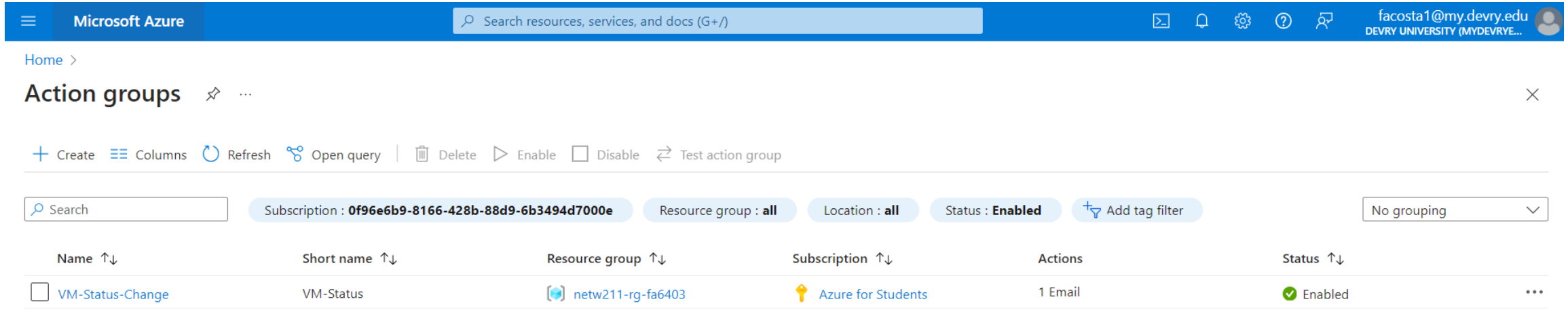
Cloud Monitoring

Cloud Monitoring

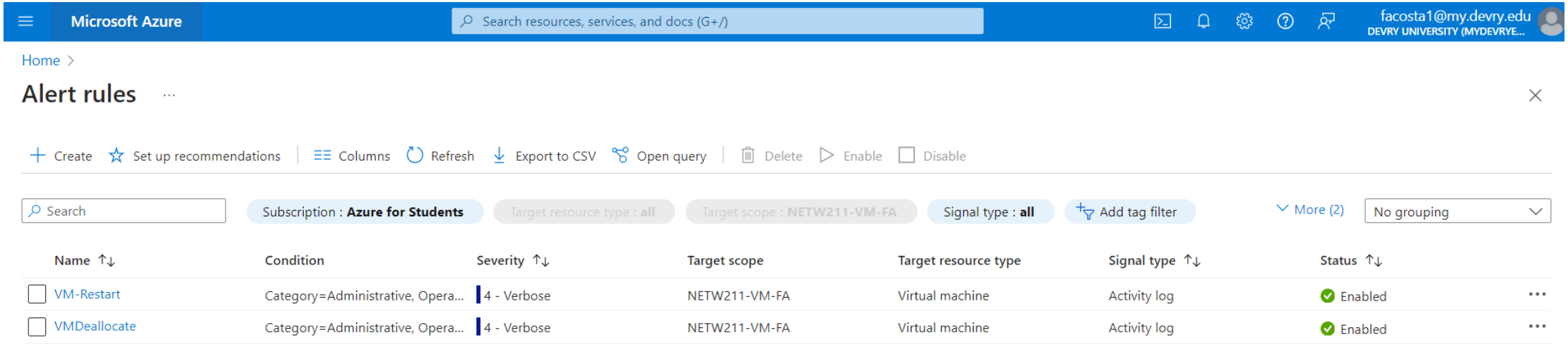
This portion of the module focused on monitoring resources, creating alert rules, and setting up notifications that are triggered when a VM event occurs within Microsoft's Azure Monitor cloud service.

Creating a VM, setting up an action group, configuring alert rules, and testing notification delivery were major steps necessary to complete this section of the project. The purpose was to create alert rules notifying designated emails when a VM is restarted or stopped.

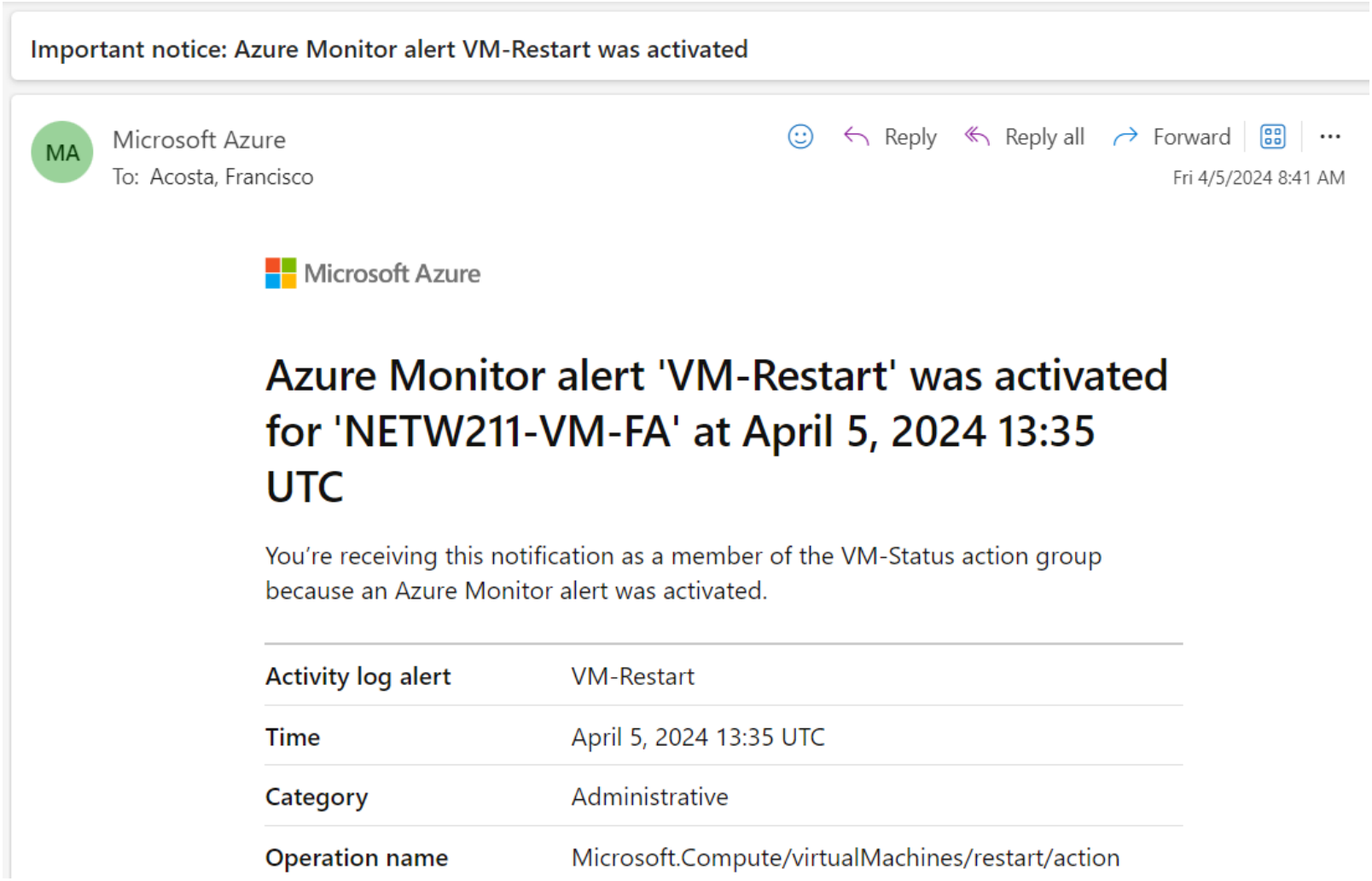
Setting up an Action Group and Notifications



Setting up Alert Rules



Testing Alerts 1 of 2



Important notice: Azure Monitor alert VM-Restart was activated

MA

Microsoft Azure

To: Acosta, Francisco

Reply | Reply all | Forward

Fri 4/5/2024 8:41 AM

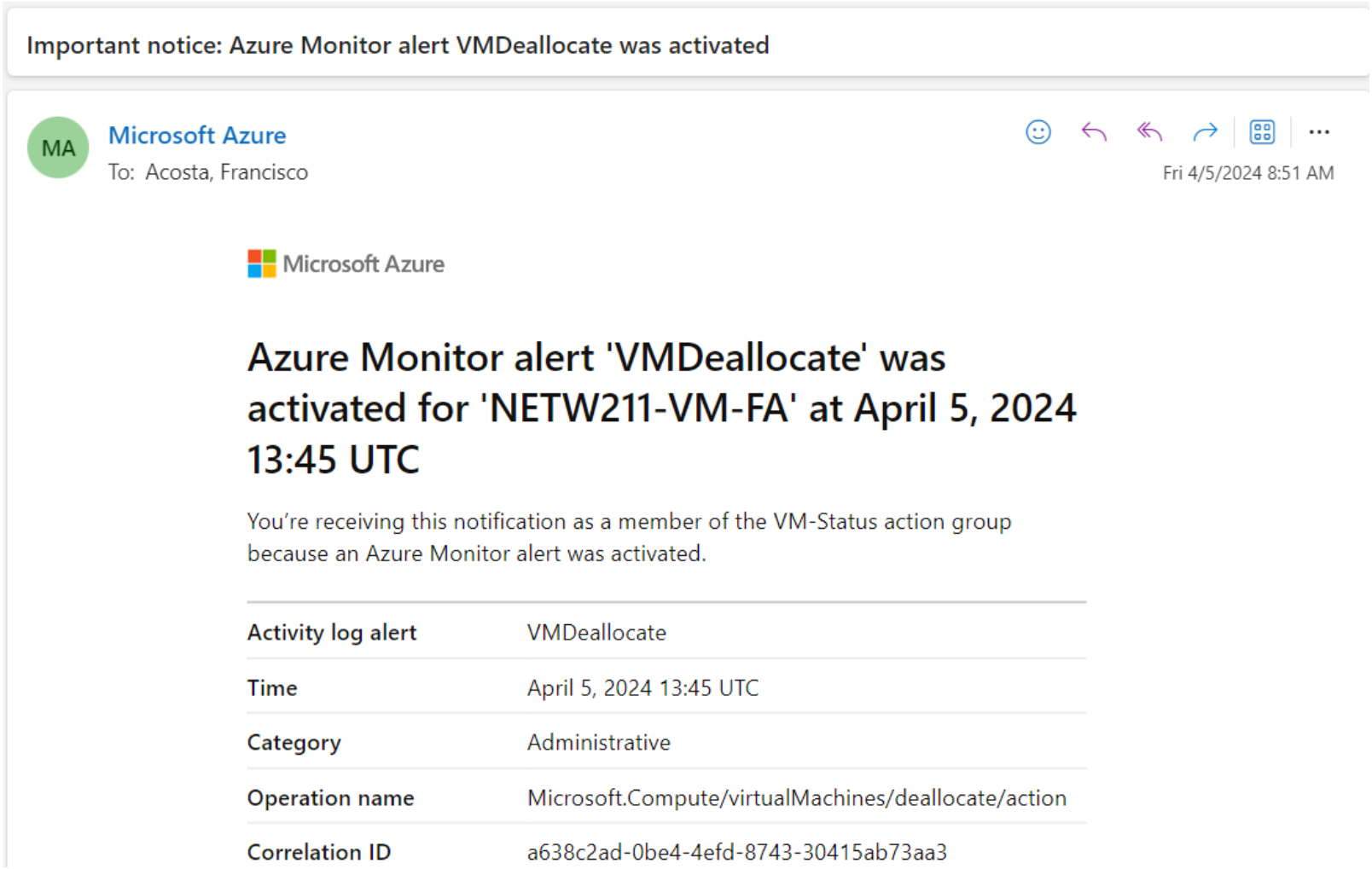
Microsoft Azure

Azure Monitor alert 'VM-Restart' was activated for 'NETW211-VM-FA' at April 5, 2024 13:35 UTC

You're receiving this notification as a member of the VM-Status action group because an Azure Monitor alert was activated.

Activity log alert	VM-Restart
Time	April 5, 2024 13:35 UTC
Category	Administrative
Operation name	Microsoft.Compute/virtualMachines/restart/action

Testing Alerts 2 of 2



Important notice: Azure Monitor alert VMDeallocate was activated

MA

Microsoft Azure

To: Acosta, Francisco

Fri 4/5/2024 8:51 AM

Microsoft Azure

Azure Monitor alert 'VMDeallocate' was activated for 'NETW211-VM-FA' at April 5, 2024 13:45 UTC

You're receiving this notification as a member of the VM-Status action group because an Azure Monitor alert was activated.

Activity log alert	VMDeallocate
Time	April 5, 2024 13:45 UTC
Category	Administrative
Operation name	Microsoft.Compute/virtualMachines/deallocate/action
Correlation ID	a638c2ad-0be4-4efd-8743-30415ab73aa3

Challenges

Challenges

Challenges are an inevitable part of any project, and this endeavor is no exception. Below are some challenges faced during this module.

- Navigating the Microsoft Azure platform
- VM instance management
- Network and virtual machine configurations
- Utilization of command prompts
- Establishing the proper configurations for a Network Security Group (NSG)
- Configuring the correct functionality of a storage account

Career Skills

Career Skills

Cloud computing is an in-demand area of expertise, and I was able to obtain a variety of skills throughout this module. Below are some key skills learned.

- Working knowledge of Microsoft Azure
- Deploying, connecting, and deleting a Virtual Machine (VM)
- Creating a VNet with subnets
- Verifying connectivity of VMs
- Connecting to a VM via the Secure Shell (SSH) protocol
- Configuring a Network Security Group (NSG)
- Creating a Blob Snapshot with versioning
- Securing a virtual machine by controlling traffic
- Setting up an action group with event triggering alerts notifications

Conclusions

Conclusions

This module highlights the importance of cloud storage solutions and the advantages they offer.

The Microsoft Azure platform was utilized to demonstrate a variety of leaned skills through practical application in real-world scenarios.

Tasks included the deployment of Virtual Machines, creation of a VNet with subnets, connectivity testing, accessing a VM via the Secure Shell (SSH) protocol, Network Security Group (NSG) configurations, management of Blob storage, and the creation of alert rules within an action group to notify designated emails of updates.

I appreciate your time and I am thankful for the opportunity to learn, as well as apply learned skills throughout this module.

References

References

Azure Virtual Network frequently asked questions.
https://docs.microsoft.com/en-us/azure/virtual-network/virtual-networks-faq

Calculator.net. IP Subnet Calculator.
https://www.calculator.net/ip-subnet-calculator.html

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DeVry University. Developing Solutions for Microsoft Azure: Blob Storage.
https://devry.percipio.com/courses/c7ef0333-8560-403f-a004-9c5c843866b0/videos/2658bbe6-ee97-438b-a376-fbb079c3b3a0

References

Microsoft. Azure Virtual Network frequently asked questions. https://docs.microsoft.com/en-us/azure/virtual-network/virtual-networks-faq

Microsoft. Access tiers for blob data. https://docs.microsoft.com/en-us/azure/storage/blobs/access-tiers-overview